CHRISTIAN MEDICAL COLLEGE VELLORE

Chromosome scale reference genome model - *Acanthamoeba polyphaga* (ATCC-30461) – T4 genotype

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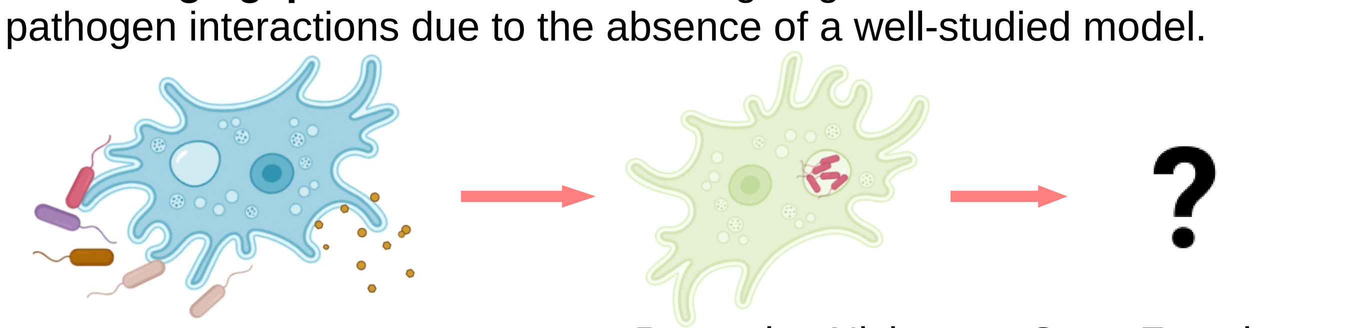
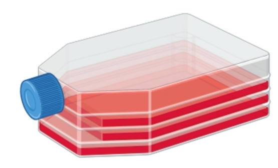
Background

- *Acanthamoeba* are ubiquitous free living protozoa, they prey on smaller microorganisms, they also serve as protective niches that support the survival and proliferation of various bacterial pathogens in natural ecosystems and also act as opportunistic pathogens.
- **Knowledge gap** - Limited understanding of genes and functions in host–pathogen interactions due to the absence of a well-studied model.

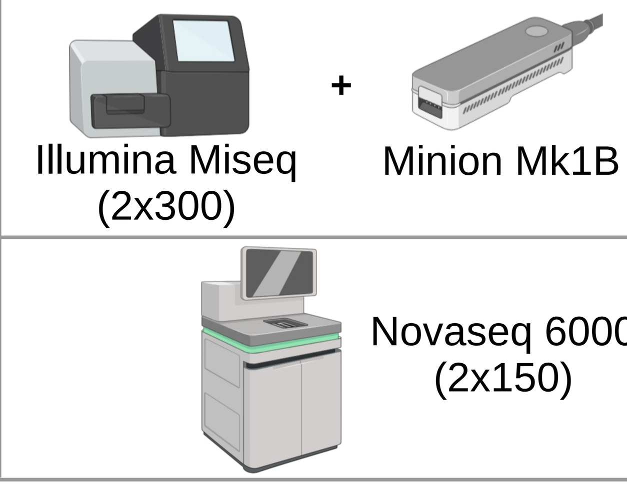


Methods

Enrichment → PYG medium → Monolayer with 3 × 10^6 cells/mL



	Extraction	Library- Preparation	Sequencing
Whole Genome Sequencing	QIAamp DNA Mini Kit	• Nextera XT kit • Rapid Barcoding kit	Illumina Miseq (2x300) + Minion Mk1B
Transcriptome Sequencing (RNA)	QIAamp viral RNA Mini Kit	Modified NEBNext RNA Ultra II kit	Novaseq 6000 (2x150)



Results

Workflow

Short reads | Long reads | Transcriptomic reads

Read QC
- *fastp / fastplong*
- Adaptor removal → Quality trimming [Q>20/30]

Genome Annotation

Genome Assembly
- *SPAdes* — hybrid – de-novo assembly → *QUAST* — Assembly stats

Contig Annotation
- *CoverM* — Contig coverage → No coverage contigs
- *BLAST* — NCBI-NT + AcanthWGS db (n= 28)
- *CAT* — NCBI-NR
- → No-Annotation / Bacteria hits
- Filtered Assembly

Gene prediction

Transcriptomics data

RNA Mapping
- Repeat Modeler
- Repeat Masking
- *HISAT2*
- *BRAKER-3* → braker.gtf
- *gffread* → Protein.faa

Protein function prediction

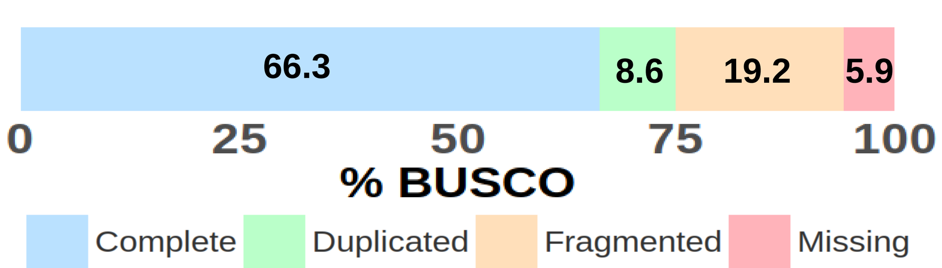
Functional Annotation

Sequence homology	Structural homology	Deep learning approach
diamond	*Foldseek*	*DeepFRI*
eggNOG, Interproscan, UniRef100, NCBI-NR	AFDB50	Gene Ontology model

→ **hybrid annotations.gtf**

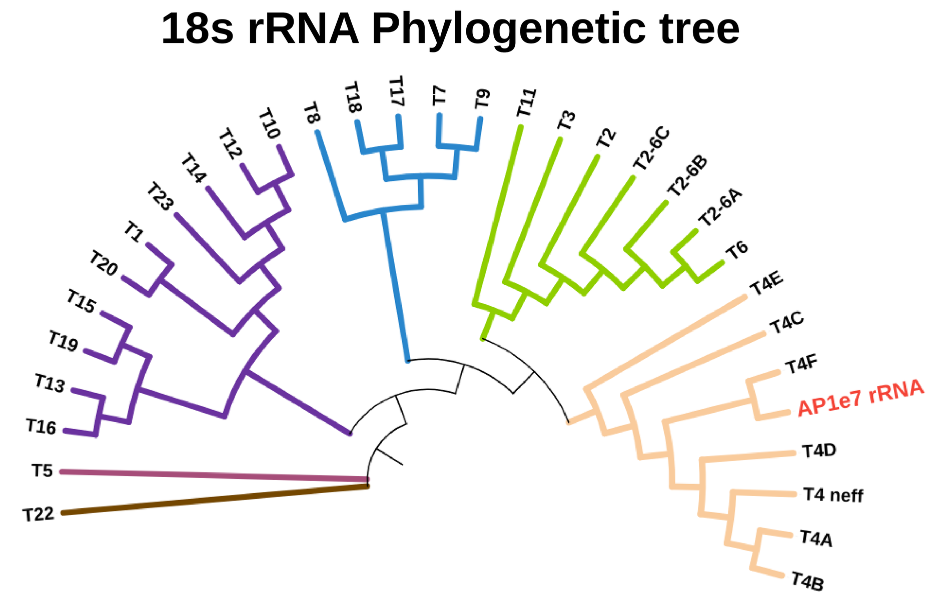
- Assembled contigs were taxonomically classified using NCBI and a custom *Acanthamoeba* WGS database (28 genomes)
- Contigs matching to *Acanthamoeba* were retained, while bacterial and unclassified contigs were excluded
- Gene prediction in BRAKER3 using RNA reads resulted in 22,067 genes and 23,998 transcripts, including 1,931 alternative transcripts
- The predicted genes were assessed for completeness in BUSCO using busco lineage eukaryota_odb10

BUSCO Assessment



- Ribosomal rRNA genes were extracted from assembled genomes using barrnap.

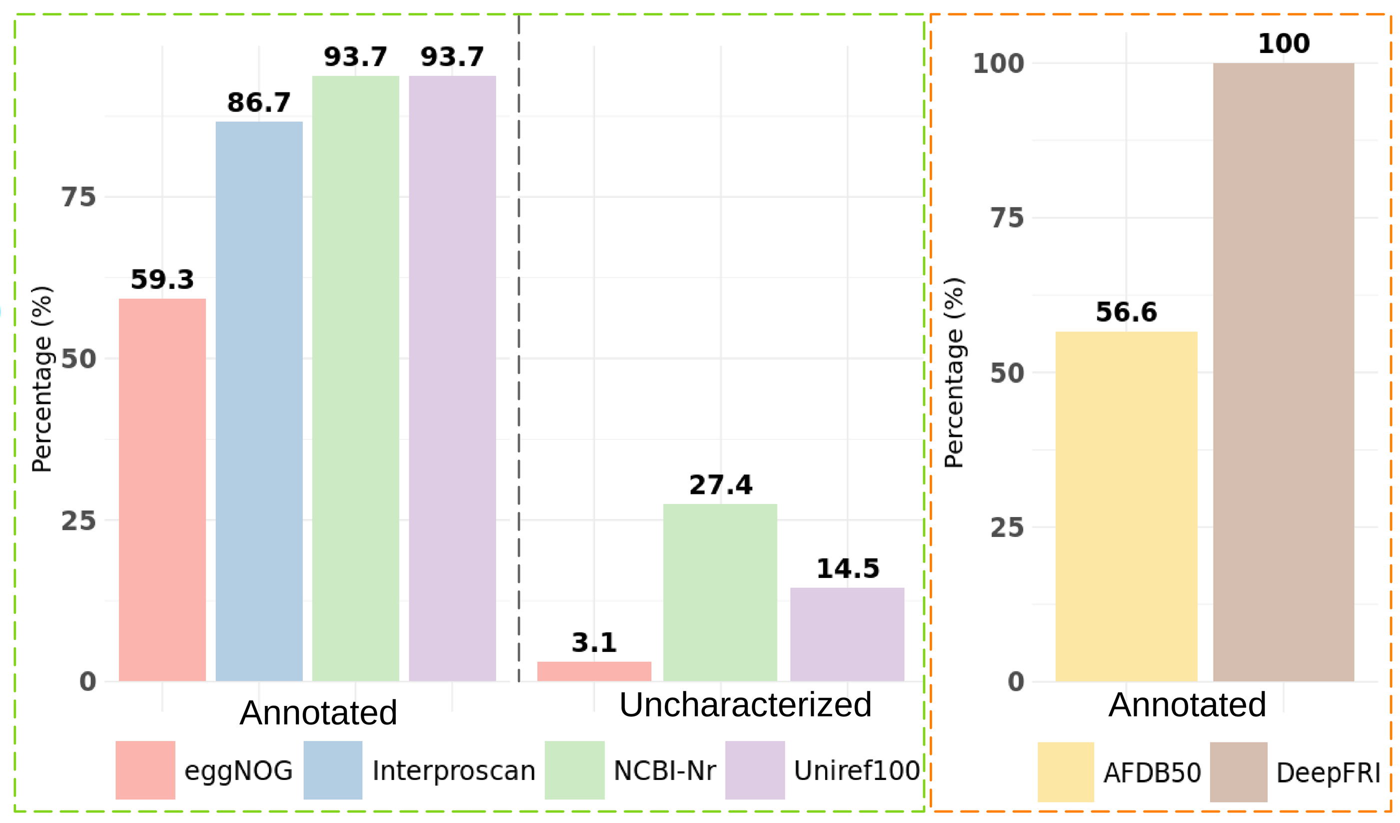
18s rRNA Phylogenetic tree



- The phylogenetic analysis in RAxML (GTR+ GAMMA+I model with 500 bootstraps) across 23 other *Acanthamoeba* genotypes revealed that ATCC-30461 clusters within the T4 genotype and is closely related to the T4F subgenotype.

- Sequence-based annotation with **NCBI nr** and **UniRef100** left 15–27% of proteins uncharacterized.
- Structural homology search with **FoldSeek** (ProsT5 3Di tokens) on **AFDB50** reduced the fraction of uncharacterized proteins to ~43%.
- **DeepFRI** graph-based structural annotation assigned **GO** terms to all proteins, achieving 100% functional annotation.

Comparison of sequence and structure driven functional annotations

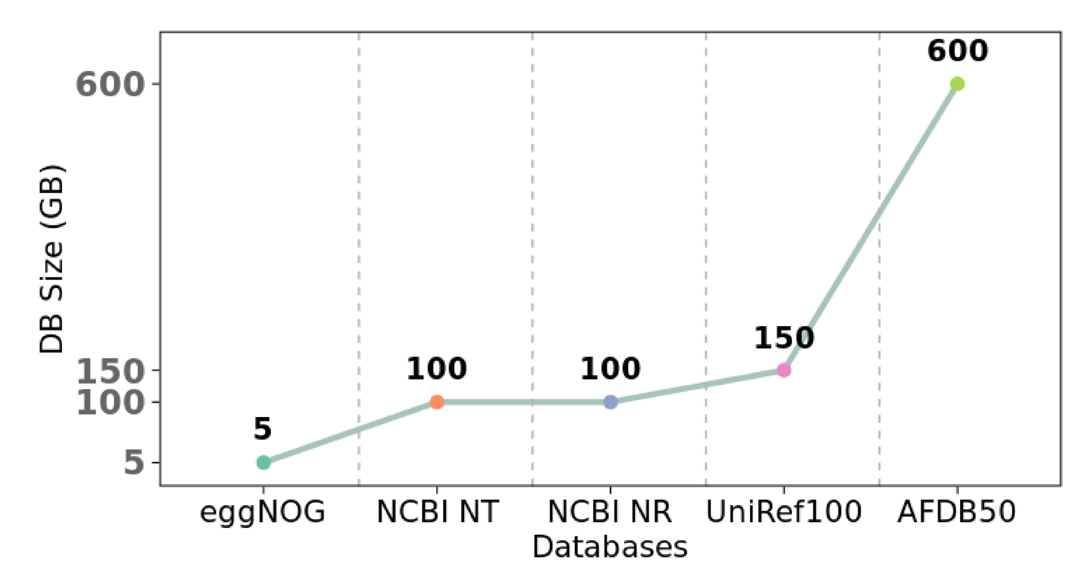


Conclusions:

- The annotated proteins of ATCC-30461 can be used as a reference for annotating other *Acanthamoeba* species and studying pathogen interactions.
- Around 40% of genes lack orthologous information in databases like eggNOG, and 14.5% are uncharacterized in UniRef100, our integrated approach provides 100% annotation for genes based on their closest known functions.
- Future work will focus on developing machine learning models that integrate sequence and structural features to achieve robust functional annotation without relying on large databases.

Limitations:

- The size of the databases hinders pipeline construction in Nextflow and Snakemake.



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For Queries and Suggestions